

Certified analytical bounds for hypersonic reachable sets

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Abstract

A method for computing *certified* reachable sets of a full six-degree-of-freedom hypersonic entry vehicle, reaching state dimensions where grid-based dynamic programming cannot, with a guarantee that is derived analytically and verified in exact arithmetic rather than produced by a numerical solver.

Open items

The problem

The reachable set of an entry vehicle (every state it can be driven to under admissible control) is what safety and threat analyses need bounded. Computing it exactly means solving a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in state dimension and capped in practice near four or five states. A rigid-body entry vehicle carries thirteen states, $x = (r, \theta, \phi, u, v, w, q_0, q_3, \omega_{1:3}) \in \mathbb{R}^{13}$, and twenty-five once ablation recession, integrated heat load and aerothermoelastic deformation are coupled. So the exact grid-HJI method quits before the real problem starts. The standard polynomial barrier with coefficients obtained from an interior-point solver has analytical *form* but inherits the solver’s accuracy, so it is not a certificate. There are then two problems to solve, not one: dimension, and the provenance of the guarantee.

The approach

Embed the hybrid flight dynamics into a Banach space of occupation measures. The nonlinear flight mechanics $\dot{x} = f_q(x, c, t)$ become *linear*: the Liouville equation is a linear constraint on a measure μ over $[0, T] \times \mathcal{X} \times U$, and $\text{supp } \mu_T$ outer-approximates the reachable set. So far this is the occupation-measure / moment-SOS program of Lasserre, Henrion and collaborators.

The new content is what happens in the thin-atmosphere limit $\varepsilon_{\text{atm}} = H_s/R_e \rightarrow 0$. Rescaled by the phugoid-to-secular time ratio (which, as in the classical Chapman–Loh variables, enters as $\sqrt{\varepsilon_{\text{atm}}}$, not ε_{atm}), the occupation-measure family *is not tight*. Mass concentrates onto the low-dimensional equilibrium-glide slow manifold $\Lambda = L \cos \sigma / [m(g - V^2/r)] = 1$. This is precisely Lions’ concentration alternative, read on the defect measure of the rescaled family.

Rather than restore compactness, we extract the profile decomposition the dichotomy hands us:

- a finite-dimensional *skeleton* $\mathcal{S}$ —the concentration component, the glide manifold as a graph over its retained coordinates;
- one *boundary-layer bubble* per skip event, the analogue of a concentrating profile: a compact impulse set $\mathcal{I}_j$ of dimension $\leq m$, carried by an internal scaling group at each atmospheric pass;
- a *compact residual* with a quantitative bound, $O(\sqrt{\varepsilon})$ in the corridor metric.

The number of bubbles is bounded *a priori*, $K \leq K_{\max}(Q_{\max})$, by the integrated heat-load budget (a trajectory can only afford so many skips). The multi-parameter separation of the three scaling groups (atmospheric, attitude, elastic) is the Bahouri–Gérard almost-orthogonality argument, transported to defect measures; it is the one genuinely new obligation and it sits on the critical path. The target statement is

$$\mathcal{R}_{\text{cert}}(w) \subseteq \underbrace{\left(\Phi_{\mathcal{S}}(\Theta) \right)}_{\text{skeleton}} \oplus \underbrace{\left(\bigoplus_{j=1}^K \mathcal{I}_j \right)}_{\text{bubbles}} \oplus \underbrace{\mathbb{B}(C_{\mathcal{R}}\sqrt{\varepsilon})}_{\text{residual}},$$

uniformly over the admissible family generated by the uncertainty set, not along a nominal trajectory, with $C_{\mathcal{R}}$ and $K_{\max}$ computable from the corridor bounds alone.

A Coupled Entry Dynamics and Polynomial Embedding

A.1 Coupled 6-DOF Entry Dynamics

The entry vehicle state comprises the coupled translational, rotational, thermal, ablative, and modal subsystems. Position is carried in Cartesian Earth-centred Earth-fixed (ECEF) components and attitude as a quaternion from the ECEF frame to the body frame. No angle is a state, so the dynamics have no coordinate singularities anywhere on the globe; geocentric latitude and longitude are read out from the position after integration.

Frames.

- $\mathcal{E}$: Earth-centred Earth-fixed (ECEF), rotating with the constant planetary rate $\vec{\omega}_E = \omega_E \hat{z}_{\mathcal{E}}$.
- $\mathcal{B}$: body frame, fixed at the instantaneous center of mass.

The attitude quaternion $\vec{q} = (q_0, q_1, q_2, q_3)^\top$ carries $\mathcal{E} \rightarrow \mathcal{B}$, with direction-cosine matrix

$$\mathbf{R} \equiv \mathbf{R}_{B/E}(\vec{q}) = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 + q_0q_1) \\ 2(q_1q_3 + q_0q_2) & 2(q_2q_3 - q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix}, \quad (\text{A.1})$$

quadratic in $\vec{q}$.

Position Kinematics. With $\vec{r} = (r_1, r_2, r_3)^\top$ the ECEF position and $\vec{V} = (u, v, w)^\top$ the planet-relative velocity resolved in $\mathcal{B}$, the kinematic equation is:

$$\dot{\vec{r}} = \mathbf{R}^\top \vec{V}. \quad (\text{A.2})$$

Translational Dynamics. Newton's law holds in the inertial frame; transporting the planet-relative velocity through $\mathcal{E}$ into $\mathcal{B}$, with inertial body rates $\vec{\omega} = (p, q, r_b)^\top$, mass m , aerodynamic force $\vec{F}_a$, thrust $\vec{F}_t$, and the planetary rate resolved in the body frame $\vec{\omega}_E^{\mathcal{B}} = \mathbf{R} \omega_E \hat{z}_{\mathcal{E}}$, the translational dynamical equations are:

$$\dot{\vec{V}}|_{\mathcal{B}} = \frac{1}{m}(\vec{F}_a + \vec{F}_t) + \mathbf{R} \vec{g}(\vec{r}) - (\vec{\omega} + \vec{\omega}_E^{\mathcal{B}}) \times \vec{V} - \vec{\omega}_E^{\mathcal{B}} \times (\vec{\omega}_E^{\mathcal{B}} \times \mathbf{R} \vec{r}). \quad (\text{A.3})$$

Gravity, including the J_2 oblateness term, in ECEF components:

$$\vec{g}(\vec{r}) = -\frac{\mu}{\|\vec{r}\|^3} \left[\left(1 + \frac{3}{2} J_2 \frac{R_e^2}{\|\vec{r}\|^2} \left(1 - 5 \frac{r_3^2}{\|\vec{r}\|^2} \right) \right) \vec{r} + 3 J_2 \frac{R_e^2}{\|\vec{r}\|^2} r_3 \hat{z}_{\mathcal{E}} \right]. \quad (\text{A.4})$$

Remark A.1. This is the gradient of the J_2 potential $\frac{\mu}{\|\vec{r}\|} [1 - \frac{1}{2} J_2 (R_e / \|\vec{r}\|)^2 (3r_3^2 / \|\vec{r}\|^2 - 1)]$; its radial and northward components reproduce the spherical form, with $r_3 / \|\vec{r}\| = \sin \phi$.

Rotational Dynamics / Euler's Equations. With angular momentum $\vec{h} = \mathbf{I} \vec{\omega}$, combined aerodynamic / reaction control system (RCS) moments $\vec{M} = \vec{M}_a + \vec{M}_{\text{RCS}}$, and the inertia tensor $\mathbf{I} = \mathbf{I}(s, \eta)$ depending on recession depth s and modal amplitudes η (so that $\dot{\mathbf{I}} \vec{\omega}$ is the variable-inertia term), the rotational dynamical equations are:

$$\mathbf{I} \dot{\vec{\omega}} = \vec{M} - \vec{\omega} \times (\mathbf{I} \vec{\omega}) - \dot{\mathbf{I}} \vec{\omega}. \quad (\text{A.5})$$

Quaternion Kinematics. The quaternion is driven by the body rate *relative to* $\mathcal{E}$, $\vec{\omega}_{B/E} = \vec{\omega} - \vec{\omega}_E^{\mathcal{B}}$. For $\vec{a} = (a_1, a_2, a_3)^\top$,

$$\dot{\vec{q}} = \frac{1}{2} \Omega(\vec{\omega} - \mathbf{R} \omega_E \hat{z}_{\mathcal{E}}) \vec{q}, \quad \Omega(\vec{a}) = \begin{pmatrix} 0 & -a_1 & -a_2 & -a_3 \\ a_1 & 0 & a_3 & -a_2 \\ a_2 & -a_3 & 0 & a_1 \\ a_3 & a_2 & -a_1 & 0 \end{pmatrix}. \quad (\text{A.6})$$

Remark A.2. The rate is **polynomial** — bilinear in $(\vec{\omega}, \vec{q})$ and cubic in $\vec{q}$ through the Earth-rate term. With this Ω and $\mathbf{R}$ of Eq. (A.1), $\dot{\mathbf{R}} = -[\vec{\omega}_{B/E} \times] \mathbf{R}$ holds identically, and $\Omega(\vec{a})$ is skew for every $\vec{a}$, so $|\vec{q}|$ is preserved.

Thermal. With the density-square-root embedding $y = \sqrt{\rho/\rho_0}$ and $V = \sqrt{u^2 + v^2 + w^2}$, the modulus lift $\dot{\chi} = c \chi \dot{T}_w$ replaces T_w with a polynomial via Varma–Savageau–Voit recasting. Stagnation-point heating (Sutton–Graves, with nose radius R_n , sea-level density ρ_0) equations are then given by:

$$\dot{Q}_{\text{tot}} = \dot{Q} = k_Q \sqrt{\rho_0/R_n} y V^3, \quad \dot{T}_w = \frac{\dot{Q} - (T_w - T_\infty)/\tau_{\text{cond}}}{C_w}. \quad (\text{A.7})$$

Ablation. With recession rate coupled to heat flux and mass loss from char-layer blow-off, the ablative equations are given by:

$$\dot{s} = \alpha_s \dot{Q}, \quad \dot{m} = -\dot{m}_{\text{abl}}. \quad (\text{A.8})$$

Modal/Structural Elasticity. With piston-theory forcing and temperature-dependent natural frequency, per mode i our modal equations are:

$$\ddot{\eta}_i + 2\zeta_i \omega_i(T) \dot{\eta}_i + \omega_i(T)^2 \eta_i = Q_i(x, \eta, \dot{\eta}). \quad (\text{A.9})$$

A.2 Polynomial Embedding via Adjunction

A priori, the dynamics contain transcendental and rational terms. We repair each of these by adjoining one variable and one polynomial defining constraint, so the entire augmented system is polynomial in the augmented state:

1. **Transcendental Incidence:** $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin(v/V)$ are inverse-trigonometric functions of body rates, we replace them with polynomial direction-cosine ratios $(u/V, v/V, w/V)$.
2. **Square-Root Speed:** Since $V = \sqrt{u^2 + v^2 + w^2}$ carries a square root, we adjoin V with the velocity magnitude constraint $V^2 = u^2 + v^2 + w^2$.
3. **Exponential Atmosphere:** Since $\rho = \rho_0 e^{-h/H_s}$ with $h = \|\vec{r}\| - R_e$ appears in $\bar{q} = \frac{1}{2} \rho V^2$, we adjoin $y = \sqrt{\rho/\rho_0}$ with rate $\dot{y} = -\frac{y}{2H_s} \frac{d}{dt} \|\vec{r}\|$, where the radial rate $\frac{d}{dt} \|\vec{r}\| = \varrho(\mathbf{R}\vec{r}) \cdot \vec{V}$ is polynomial once ϱ (item 4) is adjoined.
4. **Reciprocal Radius:** Gravity carries $\|\vec{r}\|^{-3}$, $\|\vec{r}\|^{-5}$, $\|\vec{r}\|^{-7}$, so we adjoin $\varrho = 1/\|\vec{r}\|$ with $\varrho^2 \|\vec{r}\|^2 = 1$, $\varrho > 0$, and rate $\dot{\varrho} = -\varrho^3 (\mathbf{R}\vec{r}) \cdot \vec{V}$. Gravity is then polynomial:

$$\vec{g} = -\mu \varrho^3 \left[\left(1 + \frac{3}{2} J_2 R_e^2 \varrho^2 (1 - 5 r_3^2 \varrho^2) \right) \vec{r} + 3 J_2 R_e^2 \varrho^2 r_3 \hat{z}_{\mathcal{E}} \right]. \quad (\text{A.10})$$

No trigonometric adjunctions are needed: no angle is a state, and latitude dependence enters only through $r_3 \varrho = \sin \phi$.

5. **Quaternion Norm:** $|\vec{q}|^2 = 1$ is already a polynomial invariant (skew-symmetry of Ω preserves it), no adjunction necessary.
6. **Heat / Load-Factor Square Roots:** $\dot{Q} \propto yV^3$ is already polynomial; however, $n = \|\vec{F}_a\|/(mg_0)$ requires squaring to $(\bar{q}S)^2(C_X^2 + C_Y^2 + C_Z^2) \leq (n_{\max}mg_0)^2$.

The Augmented State. Collecting all coordinates:

$$\tilde{x} = \left(\underbrace{r_1, r_2, r_3, u, v, w, q_0, q_1, q_2, q_3, p, q, r_b}_{13, \text{ core}}, \underbrace{y, V, \varrho, Q_{\text{tot}}, m, s, \chi}_{7, \text{ adjoined}}, \underbrace{\eta, \dot{\eta}}_{2n_\eta, \text{ modal}} \right). \quad (\text{A.11})$$

References